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Lab project:Data masking.

Intoduction.

- **Data masking deals with techniques that can be used in situations where data sets consisting of sensitive (confidential) information are "masked".**
- **The masked data retains its usefulness without compromising privacy and/or confidentiality.**
- **The masked data can be analyzed, shared, or disseminated without risk of disclosure.**

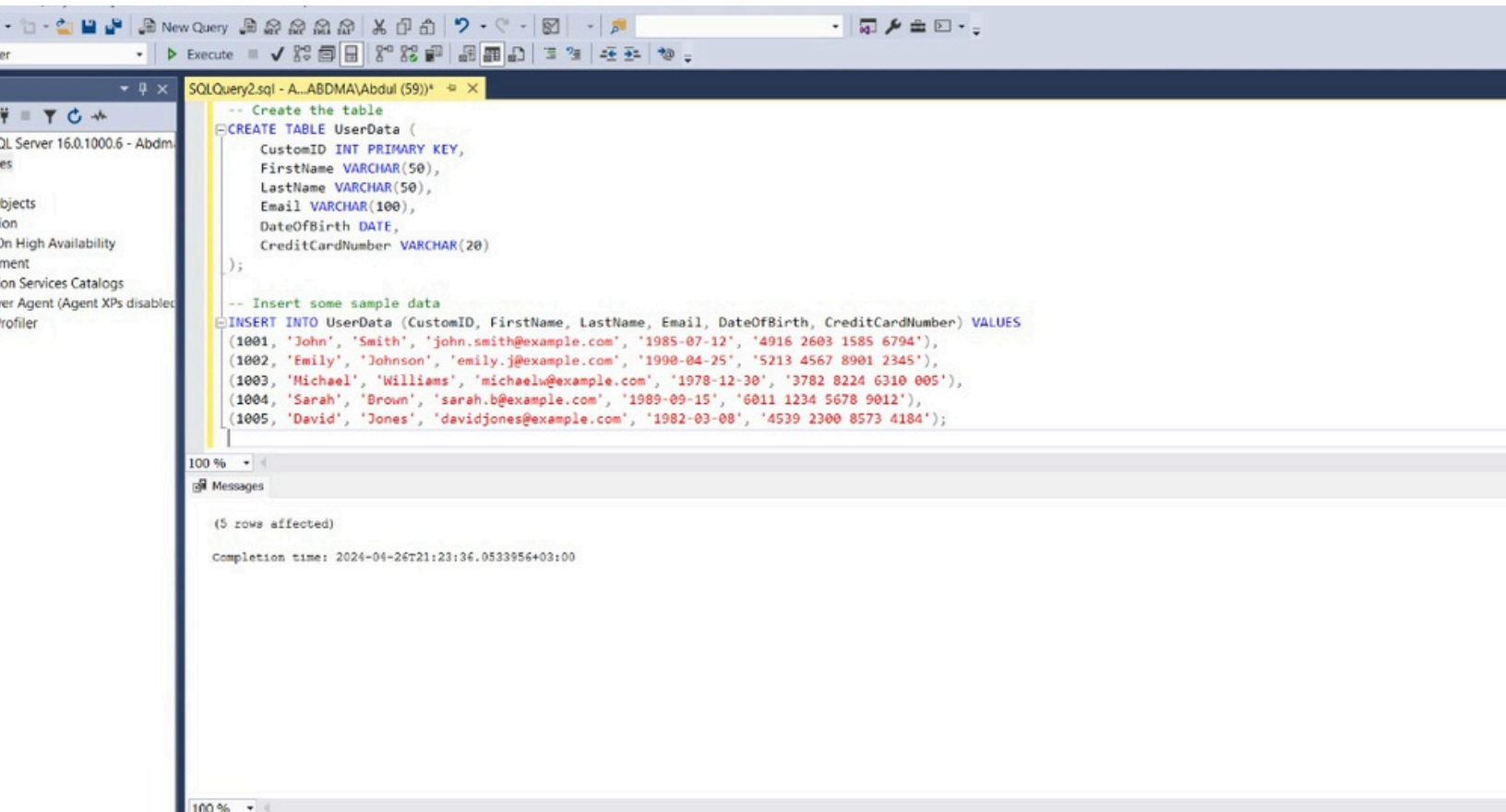
Purpose of Data Masking.

- **The primary goal of data masking is to protect sensitive data from unauthorized access while still providing a functional alternative for scenarios where the real data is not necessary.**

Best Practices:

- **Develop a comprehensive data masking policy and strategy.**
- **Classify and identify sensitive data to be masked.**
- **Choose appropriate masking techniques based on data sensitivity and use cases.**
- **Implement robust access controls and monitoring mechanisms.**
- **Test and validate the masking process to ensure data integrity and quality.**
- **Regularly review and update masking rules and techniques.**

The first step we will create a table.



The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the server hierarchy for 'SQL Server 16.0.1000.6 - Abdm...'. The right pane shows a query window titled 'SQLQuery2.sql - A...ABDMA\Abdul (59))'. The query contains the following SQL code:

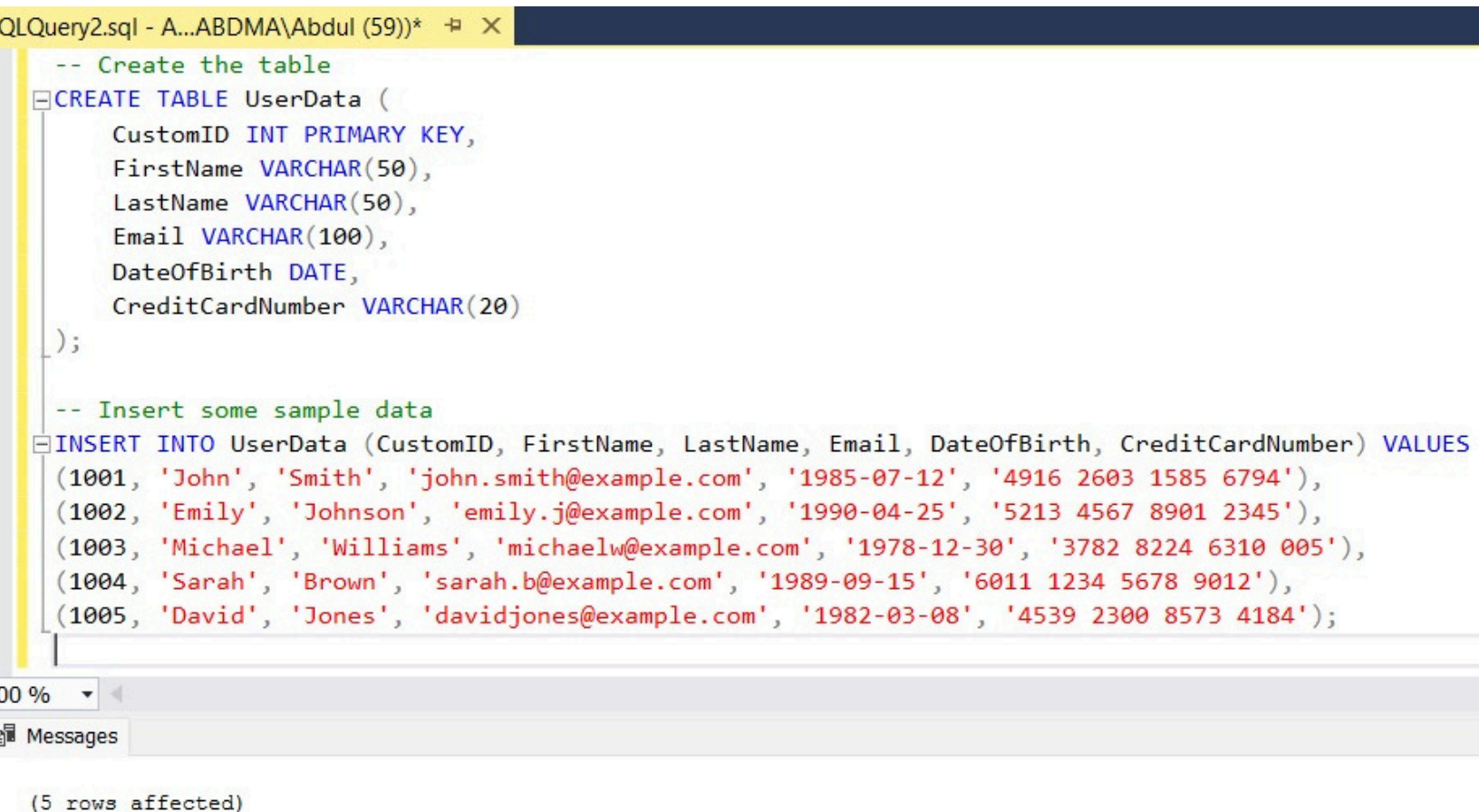
```
-- Create the table
CREATE TABLE UserData (
    CustomID INT PRIMARY KEY,
    FirstName VARCHAR(50),
    LastName VARCHAR(50),
    Email VARCHAR(100),
    DateOfBirth DATE,
    CreditCardNumber VARCHAR(20)
);

-- Insert some sample data
INSERT INTO UserData (CustomID, FirstName, LastName, Email, DateOfBirth, CreditCardNumber) VALUES
(1001, 'John', 'Smith', 'john.smith@example.com', '1985-07-12', '4916 2603 1585 6794'),
(1002, 'Emily', 'Johnson', 'emily.j@example.com', '1990-04-25', '5213 4567 8901 2345'),
(1003, 'Michael', 'Williams', 'michaelw@example.com', '1978-12-30', '3782 8224 6310 005'),
(1004, 'Sarah', 'Brown', 'sarah.b@example.com', '1989-09-15', '6011 1234 5678 9012'),
(1005, 'David', 'Jones', 'davidjones@example.com', '1982-03-08', '4539 2300 8573 4184');
```

Below the query window, the 'Messages' pane shows the execution results:

```
(5 rows affected)

Completion time: 2024-04-26T21:23:36.0533956+03:00
```



This is a close-up view of the SQL query code from the previous screenshot. It shows the full text of the 'CREATE TABLE' and 'INSERT INTO' statements. The 'Messages' pane at the bottom indicates that 5 rows were affected by the insert operation.

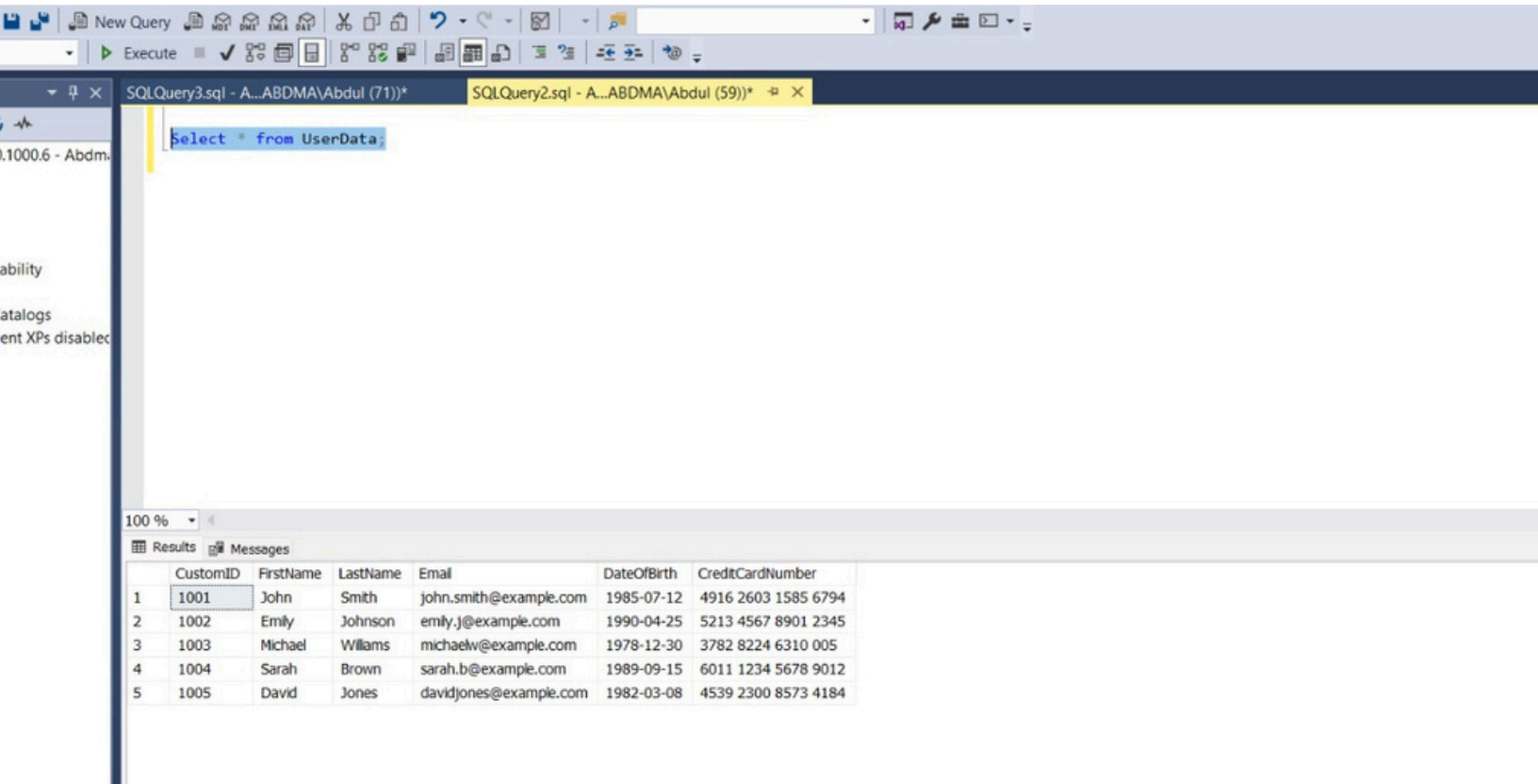
```
-- Create the table
CREATE TABLE UserData (
    CustomID INT PRIMARY KEY,
    FirstName VARCHAR(50),
    LastName VARCHAR(50),
    Email VARCHAR(100),
    DateOfBirth DATE,
    CreditCardNumber VARCHAR(20)
);

-- Insert some sample data
INSERT INTO UserData (CustomID, FirstName, LastName, Email, DateOfBirth, CreditCardNumber) VALUES
(1001, 'John', 'Smith', 'john.smith@example.com', '1985-07-12', '4916 2603 1585 6794'),
(1002, 'Emily', 'Johnson', 'emily.j@example.com', '1990-04-25', '5213 4567 8901 2345'),
(1003, 'Michael', 'Williams', 'michaelw@example.com', '1978-12-30', '3782 8224 6310 005'),
(1004, 'Sarah', 'Brown', 'sarah.b@example.com', '1989-09-15', '6011 1234 5678 9012'),
(1005, 'David', 'Jones', 'davidjones@example.com', '1982-03-08', '4539 2300 8573 4184');
```

Messages

```
(5 rows affected)
```

The table.



The screenshot shows the SQL Server Enterprise Manager interface. At the top, there's a toolbar with various icons for file operations, execution, and formatting. Below the toolbar, two query windows are open: 'SQLQuery3.sql - A...ABDMA\Abdul (71))*' and 'SQLQuery2.sql - A...ABDMA\Abdul (59))*'. The active window, 'SQLQuery3.sql', contains the SQL statement: `Select * from UserData;`. Below the query window, the 'Results' pane displays the output of the query as a table with 6 columns: CustomID, FirstName, LastName, Email, DateOfBirth, and CreditCardNumber. The table contains 5 rows of data.

	CustomID	FirstName	LastName	Email	DateOfBirth	CreditCardNumber
1	1001	John	Smith	john.smith@example.com	1985-07-12	4916 2603 1585 6794
2	1002	Emily	Johnson	emily.j@example.com	1990-04-25	5213 4567 8901 2345
3	1003	Michael	Williams	michaelw@example.com	1978-12-30	3782 8224 6310 005
4	1004	Sarah	Brown	sarah.b@example.com	1989-09-15	6011 1234 5678 9012
5	1005	David	Jones	davidjones@example.com	1982-03-08	4539 2300 8573 4184

```
Alter Table UserData  
Alter COLUMN FirstName ADD MASKED WITH (FUNCTION = 'default()');
```

100 %

Messages

Commands completed successfully.

Completion time: 2024-04-26T21:40:10.1773599+03:00

If we do execute we will see that there is no effect on the table.

```
Select * from UserData;
```

```
Alter Table UserData  
Alter COLUMN FirstName ADD MASKED WITH (FUNCTION = 'default()');
```

100 %

Results Messages

	CustomID	FirstName	LastName	Email	DateOfBirth	CreditCardNumber
1	1001	John	Smith	john.smith@example.com	1985-07-12	4916 2603 1585 6794
2	1002	Emily	Johnson	emily.j@example.com	1990-04-25	5213 4567 8901 2345
3	1003	Michael	Williams	michaelw@example.com	1978-12-30	3782 8224 6310 005
4	1004	Sarah	Brown	sarah.b@example.com	1989-09-15	6011 1234 5678 9012
5	1005	David	Jones	davidjones@example.com	1982-03-08	4539 2300 8573 4184

So we will create a new user in the database named TESTUSER, but it does not associate this user with a login from the server. This means that TESTUSER can only access the database and not the server.



The screenshot shows a SQL command being entered into a text field. The command is 'Create User TESTUSER WITHOUT LOGIN;'. The text is highlighted in blue. Below the text field, there is a status bar showing '100 %' and a 'Messages' tab. The Messages tab is active, showing the message 'Commands completed successfully.' and the completion time '2024-04-26T21:53:15.5934467+03:00'.

```
Create User TESTUSER WITHOUT LOGIN;
```

Commands completed successfully.

Completion time: 2024-04-26T21:53:15.5934467+03:00

Then we will grants TESTUSER the permission to select (i.e., read) data from the dbo schema. The dbo schema is the default schema in SQL Server and contains most of the system objects.

```
Create User TESTUSER WITHOUT LOGIN;
```

```
GRANT SELECT on SCHEMA::dbo to TESTUSER;
```

100 %

Messages

Commands completed successfully.

Completion time: 2024-04-26T21:55:26.2920443+03:00

This command switches the execution context to the TESTUSER user. Any subsequent commands executed after this will be performed as if the TESTUSER were executing them.

```
GRANT SELECT on SCHEMA::dbo to TESTUSER;

EXECUTE AS USER = 'TESTUSER';

Select * from UserData;
```

100 %

Results Messages

	CustomID	FirstName	LastName	Email	DateOfBirth	CreditCardNumber
1	1001	xxxx	Smith	john.smith@example.com	1985-07-12	4916 2603 1585 6794
2	1002	xxxx	Johnson	emily.j@example.com	1990-04-25	5213 4567 8901 2345
3	1003	xxxx	Williams	michaelw@example.com	1978-12-30	3782 8224 6310 005
4	1004	xxxx	Brown	sarah.b@example.com	1989-09-15	6011 1234 5678 9012
5	1005	xxxx	Jones	davidjones@example.com	1982-03-08	4539 2300 8573 4184

As we see the masking has been performed on the FirstName column.

Add the masking.

Adds a mask to the CreditCardNumber column. The partial function is used to mask all characters except the last four, which will be replaced by 'x's.

SQLQuery1.sql - A...ABDIMA\Abdul (71))
SQLQuery2.sql - A...ABDIMA\Abdul (59))

```
Alter Table UserData  
Alter COLUMN CreditCardNumber ADD MASKED WITH (FUNCTION = 'partial(0,"xxxxxxxxxxx",4)');
```

100 %

Messages

Commands completed successfully.

Completion time: 2024-04-26T22:20:19.0904745+03:00

```
GRANT SELECT on SCHEMA::dbo to TESTUSER;
```

```
EXECUTE AS USER = 'TESTUSER';
```

```
Select * from UserData;
```

ablec

100 %

Results Messages

	CustomID	FirstName	LastName	Email	DateOfBirth	CreditCardNumber
1	1001	xxxx	Smith	john.smith@example.com	1985-07-12	xxxxxxxxxxx6794
2	1002	xxxx	Johnson	emily.j@example.com	1990-04-25	xxxxxxxxxxx2345
3	1003	xxxx	Williams	michaelw@example.com	1978-12-30	xxxxxxxxxxx 005
4	1004	xxxx	Brown	sarah.b@example.com	1989-09-15	xxxxxxxxxxx9012
5	1005	xxxx	Jones	davidjones@example.com	1982-03-08	xxxxxxxxxxx4184

Adds a mask to the CreditCardNumber column. The partial function is used to mask all characters except the first four, which will be replaced by 'x's.

```
Alter Table UserData
Alter COLUMN CreditCardNumber ADD MASKED WITH (FUNCTION = 'partial(4,"xxxxxxxxxxxx",0)');
```

Messages

Commands completed successfully.

Completion time: 2024-04-26T22:50:32.9054524+03:00

```
GRANT SELECT on SCHEMA::dbo to TESTUSER;

EXECUTE AS USER = 'TESTUSER';

Select * From UserData;
```

100 %

Results Messages

	CustomID	FirstName	LastName	Email	DateOfBirth	CreditCardNumber
1	1001	xxxx	Smith	jXXX@XXXX.com	1985-07-12	4916xxxxxxxxxxxx
2	1002	xxxx	Johnson	eXXX@XXXX.com	1990-04-25	5213xxxxxxxxxxxx
3	1003	xxxx	Williams	mXXX@XXXX.com	1978-12-30	3782xxxxxxxxxxxx
4	1004	xxxx	Brown	sXXX@XXXX.com	1989-09-15	6011xxxxxxxxxxxx
5	1005	xxxx	Jones	dXXX@XXXX.com	1982-03-08	4539xxxxxxxxxxxx

Adds a mask to the CreditCardNumber column. The partial function is used to replace the entire value with 'x's.

```
Alter Table UserData  
Alter COLUMN CreditCardNumber ADD MASKED WITH (FUNCTION = 'partial(0,"xxxxxxxxxxxxxxxx",0)');
```

0 %

Messages

Commands completed successfully.

Completion time: 2024-04-26T23:06:08.2549101+03:00

```
EXECUTE AS USER = 'TESTUSER';
```

```
Select * From UserData;
```

100 %

Results Messages

	CustomID	FirstName	LastName	Email	DateOfBirth	CreditCardNumber
1	1001	xxxx	Smith	jXXX@XXX.com	1985-07-12	xxxxxxxxxxxxxxxx
2	1002	xxxx	Johnson	eXXX@XXX.com	1990-04-25	xxxxxxxxxxxxxxxx
3	1003	xxxx	Williams	mXXX@XXX.com	1978-12-30	xxxxxxxxxxxxxxxx
4	1004	xxxx	Brown	sXXX@XXX.com	1989-09-15	xxxxxxxxxxxxxxxx
5	1005	xxxx	Jones	dXXX@XXX.com	1982-03-08	xxxxxxxxxxxxxxxx

Adds a mask to the Email column. The email() function is used to mask email addresses.

```
Alter Table UserData  
Alter COLUMN Email ADD MASKED WITH (FUNCTION = 'email()');
```

100 %

Messages

Commands completed successfully.

Completion time: 2024-04-26T22:30:31.8276253+03:00

```
EXECUTE AS USER = 'TESTUSER';  
Select * From UserData;
```

100 %

	CustomID	FirstName	LastName	Email	DateOfBirth	CreditCardNumber
1	1001	xxxx	Smith	jXXX@XXXX.com	1985-07-12	xxxxxxxxxxxx6794
2	1002	xxxx	Johnson	eXXX@XXXX.com	1990-04-25	xxxxxxxxxxxx2345
3	1003	xxxx	Williams	mXXX@XXXX.com	1978-12-30	xxxxxxxxxxxx 005
4	1004	xxxx	Brown	sXXX@XXXX.com	1989-09-15	xxxxxxxxxxxx9012
5	1005	xxxx	Jones	dXXX@XXXX.com	1982-03-08	xxxxxxxxxxxx4184

Remove the masking.

Here we will Removes the masking applied to the CreditCardNumber column.

This command effectively removes any masking applied to the CreditCardNumber column, meaning the original values will be displayed as they are without any obscuring or masking

```
Alter Table UserData  
Alter COLUMN CreditCardNumber DROP MASKED ;
```

0 %

Messages

Commands completed successfully.

Completion time: 2024-04-26T23:17:58.2062688+03:00

```
Select * From UserData;
```

100 %

	CustomID	FirstName	LastName	Email	DateOfBirth	CreditCardNumber
1	1001	xxxx	Smith	jXXX@XXXX.com	1985-07-12	4916 2603 1585 6794
2	1002	xxxx	Johnson	eXXX@XXXX.com	1990-04-25	5213 4567 8901 2345
3	1003	xxxx	Williams	mXXX@XXXX.com	1978-12-30	3782 8224 6310 005
4	1004	xxxx	Brown	sXXX@XXXX.com	1989-09-15	6011 1234 5678 9012
5	1005	xxxx	Jones	dXXX@XXXX.com	1982-03-08	4539 2300 8573 4184

This command will grant the UNMASK permission to the user TESTUSER. Once this permission has been granted, the user TESTUSER will be able to view the unmasked data in any table that has been masked using the MASKED function.

GRANT UNMASK to TESTUSER;

0 %

Messages

Commands completed successfully.

Completion time: 2024-04-26T23:22:00.0575211+03:00

Create User TESTUSER WITHOUT LOGIN;

GRANT SELECT on SCHEMA::dbo to TESTUSER;

EXECUTE AS USER = 'TESTUSER';

Select * From UserData;

100 %

Results Messages

	CustomID	FirstName	LastName	Email	DateOfBirth	CreditCardNumber
1	1001	John	Smith	john.smith@example.com	1985-07-12	4916 2603 1585 6794
2	1002	Emily	Johnson	emily.j@example.com	1990-04-25	5213 4567 8901 2345
3	1003	Michael	Williams	michaelw@example.com	1978-12-30	3782 8224 6310 005
4	1004	Sarah	Brown	sarah.b@example.com	1989-09-15	6011 1234 5678 9012
5	1005	David	Jones	davidjones@example.com	1982-03-08	4539 2300 8573 4184